

WPS18

CRSALE010

22 September 2022

a) We model the internal resistance as a resistor so we use Ohm's Law:

$$I = \frac{|\Delta V|}{R}$$
$$\Rightarrow 18 = \frac{9}{R} \Rightarrow R = 0.5 \Omega$$

b) We use $P = I \Delta V$

$$= 18(9)$$
$$= 162 \text{ W}$$

c) All of the power is dissipated by the internal resistance, so 162 J is dissipated every second.

d) We get $\Delta V_{\text{sys}} = 0$

$$\Rightarrow \text{emf} - \Delta V_{\text{internal}} - \Delta V_{\text{resistor}} = 0$$
$$\Rightarrow \text{emf} = \Delta V_{\text{int}} + \Delta V_{\text{res}}$$

Both are resistors, so $\Delta V = IR$

$$\Rightarrow \text{emf} = I(R_{\text{int}} + R_{\text{res}})$$

$$\Rightarrow I = \frac{\text{emf}}{R_{\text{int}} + R_{\text{res}}} = \frac{9}{10 + 0.5} = 0.857 \text{ A}$$

e) We use $P = I^2 R$

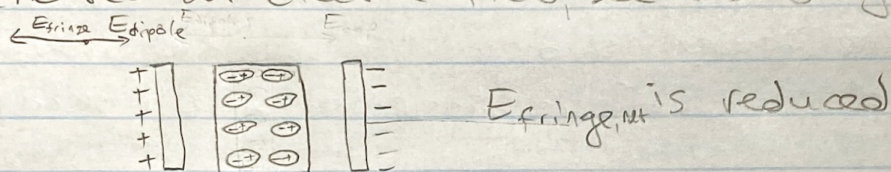
$$= 0.857^2 (10)$$
$$= 7.35 \text{ W}$$

f) This reads $\Delta V_{\text{resistor}}$

$$= I R_{\text{res}}$$
$$= 0.857(10)$$
$$= 8.57 \text{ V}$$

2a) We use $Q = C|\Delta V|$ where Q is the max charge on the plate.
 so $Q = C|\Delta V|$, but for the final state of equilibrium, $E = 0$ so $\text{emf} = \Delta V_{\text{cap}}$, $\therefore Q = \text{emf} \cdot C$

b) Current stops flowing when $E_{\text{wire}} = E_{\text{cap}}$ due to the charges on the capacitor.
 Inserting a dielectric creates an opposing electric field to E_{cap} (dipoles form which creates an electric field, see the diagram below)



This means E_{net} at the capacitor becomes less, so $E_{\text{wire}} > E_{\text{cap}}$ allowing current to flow.

c) Since $E_{\text{new}} = E_{\text{charge}} - E_{\text{dipole}}$

$$= E_{\text{charge}} - \frac{E_{\text{charge}}}{k}$$

$$= \frac{E_{\text{old}}}{k}$$

Thus $\Delta V_{\text{cap, new}} = E_s = \frac{E_{\text{old}}}{k} s = \frac{\Delta V_{\text{cap, old}}}{k}$

Now, by the Loop Rule, $\text{emf} = \Delta V_{\text{res}} + \Delta V_{\text{cap, new}}$

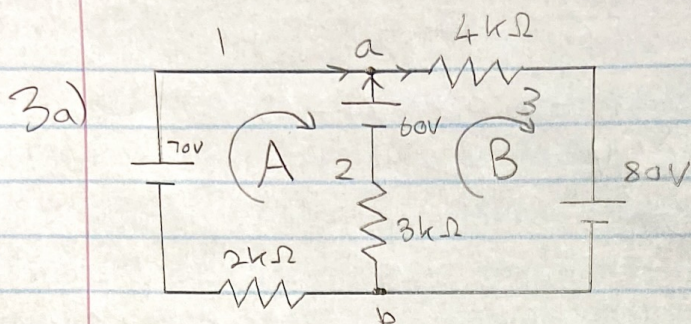
So $\text{emf} = IR + \frac{\text{emf}}{k}$ ($\Delta V_{\text{cap, old}} = \text{emf}$)

$\Rightarrow \text{emf} \left(1 - \frac{1}{k}\right) = IR$

$\Rightarrow I = \frac{\text{emf} \left(1 - \frac{1}{k}\right)}{R}$

d) $Q = C|\Delta V|$ but now the capacitance has changed. $C_{\text{new}} = \frac{Q}{|\Delta V|} = \frac{Q}{E_s} = \frac{Q}{\frac{Q/A \cdot 1/k \cdot s}} = \frac{Qk}{E_{\text{old}}}$ but $\frac{Q}{E_{\text{old}}} = C_{\text{old}}$, so

$C_{\text{new}} = C_{\text{old}} k \Rightarrow Q_{\text{new}} = C_{\text{new}} |\Delta V| = C_{\text{old}} k (\text{emf}) = Ck \text{emf}$



Using the node rule at (a):

$$I_1 + I_2 = I_3$$

Using the loop rule for A:

$$+70 - 60 + I_2(3 \times 10^3) - I_1(2 \times 10^3) = 0$$

Using the loop rule for B:

$$+60 - I_3(4 \times 10^3) - 80 - I_2(3 \times 10^3) = 0$$

Using these equations yields:

$$I_1 = 0.000384 \text{ A}$$

$$I_2 = -0.00308 \text{ A} \Rightarrow 0.00308 \text{ A in the other direction}$$

$$I_3 = -0.00269 \text{ A} \Rightarrow 0.00269 \text{ A in the other direction}$$

b) $\Delta V_{BA} = \Delta V_{\text{patt}} + \Delta V_{\text{res}}$

$$= 60 + (0.00308)(3 \times 10^3) \text{ Note up is against current, so } \Delta V_{\text{res}} > 0$$

$$= 69.24 \text{ V}$$

c) Point (a). $\Delta V_{BA} > 0$ (change in potential from B to A) meaning $V_A > V_B$. Current also flows from high to low potential and flows from A to B

4. Using the node rule (where 3 wires meet):

$$5.0 + I_1 = I_2$$

$$\text{So } I_2 = I_1 + 5$$

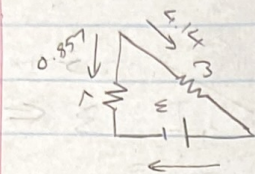
Using the loop rule for the left loop:

$$+16 - 5(2) - I_2(7) = 0$$

$$\Rightarrow I_2 = 0.857 \text{ A}$$

$$\therefore I_1 = -4.14 \text{ A} \Rightarrow 4.14 \text{ A in the other direction}$$

Using the loop rule for the right loop:



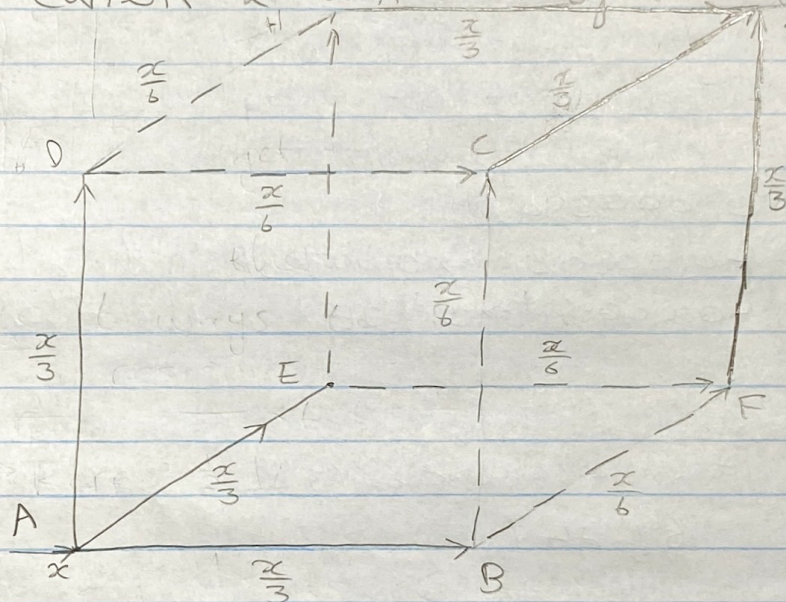
$$3(4.14) - \varepsilon - 7(0.857) = 0$$

$$\Rightarrow \varepsilon = 6.42 \text{ V}$$

So current flows to the left and $\varepsilon = 6.42 \text{ V}$
 Finally, $\Delta V = 4.14(3) = 12.4 \text{ V}$

5a) The cube looks as follows:

Consider current x at 'A'. Then, by node rule, we get:



Now for $\Delta V_{ABG} = V$, we get $V = \frac{x}{3}(1) + \frac{x}{6}(1) + (\frac{x}{3})(1)$
 for each path. So

$$V = \frac{5x}{6}$$

Meaning: $R = \frac{|\Delta V|}{I}$

$$= \frac{\frac{5x}{6}}{\frac{5x}{6}}$$

$$= \frac{5}{6} \Omega$$